Reg. No. | | | | | | | | | | | [14AE63]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE–2019

PROPULSION-II

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions selecting at least two full questions from each part.

PART - A

1. **a.** Explain the principle of thrust generation in a rocket engine. **04**

b. Derive the rocket equation. Illustrate with a diagram. **10**

2. **a.** Discuss how the design altitude is determined for optimization of thrust over flight trajectory. **04**

b. Explain the following: **06**

i. Characteristic velocity

ii. Thrust coefficient

c. A liquid propellant rocket engine operates with a chamber pressure of 20 bar and develops a 7000 N thrust at an altitude where ambient pressure is 10 millibar. The nozzle exit pressure is 50 millibar and the throat area is 1.948×10^{-3} m^2. The propellant flow rate is 3 kg/s. Calculate **04**

i. Jet velocity

ii. Thrust coefficient

3. **a.** Estimate the heat of combustion of a stoichiometric methane-air mixture. The pressure is 1 atm and the initial reactant temperature is 298 K. Assume enthalpy of formation at 298 K: **04**

For CO_2= -393.52 kJ/mol

For H_2O (vapor)= -241.83 kJ/mol

For CH_4= -74.87 kJ/mol

b. Explain

i. heat of formation **05**

ii. adiabatic flame temperature. **05**

4. **a.** Derive the expression for mass flow rate through a nozzle. **07**

b. Why is Hydrogen preferred to Hydrocarbon fuels as a rocket propellant? **02**

c. Discuss how the nozzle jet velocity can be increased. **05**

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PART - B

5. a. List the desirable properties of solid propellants. **06**

b. What is linear burning rate of solid propellant? **04**

c. The linear burning rate of a solid propellant grain is 20 mm/s when the chamber pressure is 80 bar and 40 mm/s when the chamber pressure is 200 bar. Calculate the thermal coefficient and pressure exponent of the linear burning rate. **04**

6. a. Describe the regenerative cooling method used in liquid propellant rockets. **08**

b. Sketch and label neatly the gas generator cycle. **06**

7. a. Write a note on electrothermal propulsion. **04**

b. Explain the challenges of cryogenic engines. **10**

8. a. What is the optimum number of stages for a launch vehicle? Justify. **05**

b. A two stage rocket is used to put up a satellite of 40 kg mass in a Low Earth Orbit. The details are given below: **09**

Stage	V_e (m/s)	Structure mass (kg)	Propellant mass (kg)
1	2200	1500	9000
2	2400	550	3500

Calculate the Mass Ratio for each stage and the total velocity increment of the rocket. Assume negligible drag and gravitational effects on the vehicle.